# Statistics, Counting & Probability

## Section 1: Expected Value & Probability Distributions

**1** A charity raffle sells $1,000$ tickets at $5$ each. There is one grand prize of $1,000$ and two second prizes of $500$ each. What is the expected net gain for a person purchasing one ticket?

**a.** -$3.00

**b.** -$2.00

**c.** -$1.50

**d.** $0.00

**2** A random variable $X$ has probability distribution: $P(X=1)=0.2, P(X=2)=0.5, P(X=3)=0.3$. What is the expected value $E(X)$?

**a.** $2.0$

**b.** $2.1$

**c.** $2.2$

**d.** $2.3$

**3** If $E(X)=4$ and $E(X^2)=25$, what is the variance $\text{Var}(X)$?

**a.** $3$

**b.** $5$

**c.** $9$

**d.** $16$

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## Section 2: Normal Distribution & Z-Scores

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**4** The scores on a standardized exam are normally distributed with a mean of $500$ and a standard deviation of $100$. Approximately what percentage of test takers score between $400$ and $600$?

**a.** 50%

**b.** 68%

**c.** 95%

**d.** 99.7%

**5** A student scores $84$ on a test with mean $72$ and standard deviation $8$. What is the student's $z$-score?

**a.** $1.25$

**b.** $1.50$

**c.** $1.75$

**d.** $2.00$

**6** In a normal distribution with mean $50$ and standard deviation $5$, what value corresponds to a $z$-score of $-1.6$?

**a.** 42

**b.** 44

**c.** 46

**d.** 48

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## Section 3: Conditional Probability & Bayes' Rule

**7** Given $P(A) = 0.6, P(B) = 0.5,$ and $P(A \cap B) = 0.3$, what is the conditional probability $P(A \mid B)$?

**a.** $0.50$

**b.** $0.60$

**c.** $0.75$

**d.** $0.30$

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**8** If two events $A$ and $B$ are mutually exclusive with $P(A) = 0.35$ and $P(B) = 0.45$, what is $P(A \cup B)$?

**a.** 0.70

**b.** 0.80

**c.** 0.90

**d.** 0.1575

**9** A medical screening test is 95% accurate. If 2% of a population has the disease, what is the probability that a randomly chosen person has the disease AND tests positive?

**a.** 0.019

**b.** 0.020

**c.** 0.095

**d.** 0.95

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## Section 4: Combinatorial Counting & Pigeonhole Principle

**10** In how many ways can 6 students be seated in a circle?

**a.** 120

**b.** 720

**c.** 60

**d.** 24

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**11** What is the minimum number of socks that must be drawn from a drawer containing $10$ black socks and $10$ white socks to guarantee getting at least one matching pair?

**a.** $2$

**b.** $3$

**c.** $4$

**d.** $11$

**12** How many distinct permutations can be formed using all the letters of the word 'STATISTICS'?

**a.** $50,400$

**b.** $100,800$

**c.** $302,400$

**d.** $3,628,800$

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